ACADEMIC TECHNICAL DOCUMENTATION • MINOR PROJECT

MONO TIME — Technologies, Frameworks & Software Stack

A Comprehensive Inventory of Languages, Graphics Engines, Microservices, LLMs & Databases

Author / Developer: Project Student Team | **Evaluation:** Minor Project Viva | **Architecture:** Next.js + FastAPI + Three.js + Gemini

1. Technology Stack Master Classification

Mono Time integrates bleeding-edge modern web technologies across client-side WebGL rendering, asynchronous REST micro-gateways, relational transaction audit ledgers, and autonomous multimodal AI orchestration.

Layer / Subsystem	Primary Technology	Version / Engine	Role & Key Benefit
Frontend Core	Next.js (App Router)	13.5.6 / React 18.2	Server-side rendering, code-splitting, zero-layout-shift hydration
Type Safety	TypeScript	v5.3.3 (Strict)	Complete compile-time safety across products, orders & AI message payloads
Styling Engine	Tailwind CSS	v3.4.0	Custom Swiss-horology tokens (Emerald Malachite, Obsidian Noir, Platinum)
3D Graphics & Physics	Three.js (WebGL)	v0.186.0	Procedural 3D luxury chronometer, 72-facet bezel, 8Hz tourbillon escapement
Backend Gateway	FastAPI (Python)	v0.110+ (Py 3.11)	Asynchronous ASGI endpoints, Swagger OpenAPI docs, high-concurrency routing
ASGI Server	Uvicorn	v0.29+ / uvloop	Listens on 0.0.0.0:8000 for LAN, mobile hotspot, and cloud tunnels
AI LLM Engine	Google Gemini 1.5 Flash	Generative AI API	Natural language concierge, wrist proportions analysis, RAG factual grounding
Agent Graph Router	LangGraph Paradigm	Custom StateGraph	Multi-agent routing: Supervisor, Stylist, RAG Knowledge Search, Concierge Desk
Relational Database	SQLite 3 + SQLAlchemy	v2.0 ORM	Embedded ACID-compliant persistence for 15 watches & checkout_orders audit
State Management	React Context + Zustand	React 18 Hooks	Centralized state for cart, search index, customer/admin modes & chat timeline
Automated Testing	Pytest + TestClient	v9.1.1 (100% Pass)	8 integration tests verifying product filter, exit-intent observer & order reflection

2. Deep Dive: Client-Side Stack & Graphics Engine

• **Next.js 14 / 13.5 App Router Architecture:** Employs modern React Server Components and client-side boundary isolation ('use client'). Static pages (/ and /merchant) are pre-rendered at build time with 0 type errors, achieving sub-second initial paint times.

• **Three.js Procedural WebGL Chronometer:** Eliminates external 15MB 3D asset downloads. The entire watch case, 72-facet diamond-cut coin bezel, 1024x1024 radial guilloché dial, applied gold baton indices, and 5-link Jubilee bracelet are computed purely with mathematical trigonometric primitives. Parallax cursor and touchmove tracking use slerp dampening.

• **Lucide React & UI Iconography:** Tree-shakable SVG icons (Watch, Sparkles, BarChart3, ShoppingBag, ShieldCheck, Ruler, Truck, RefreshCw) styled with emerald glow accents.

• **Responsive Viewport & Touch Optimization:** Configured viewport scaling ('width=device-width, initial-scale=1, maximum-scale=5') and dynamic mobile camera distance (z=32 on smartphones vs z=24 on desktop) ensuring pristine rendering on mobile screens.

3. Deep Dive: Backend REST Services & AI Multi-Agent Architecture

• **FastAPI Asynchronous Gateway:** Lightweight Python microservice handling catalog retrieval, RAG semantic search, multimodal exit-intent observation, and checkout verification. Provides auto-generated interactive OpenAPI docs at /docs.

• **Google Gemini 1.5 Flash Integration:** Connected via google-generativeai client library. Grounded strictly with local SQLite product specifications to eliminate hallucinations, delivering accurate wrist sizing advice, water resistance ratings, and sartorial pairings.

• **LangGraph-Inspired Multi-Agent Intent Router:** User inquiries are dynamically triaged by a Supervisor Agent into Stylist recommendations, RAG catalog retrieval, or Concierge Desk dispatch.

• **SQLAlchemy Relational ORM & SQLite Persistence:** Database tables ('products' and 'checkout_orders') maintain transactional integrity. Every customer acquisition submitted through Mock Stripe is permanently stored with order ID, customer details, discount code, and line items.

4. Networking, Tunneling & Deployment Readiness

Environment	Tool / Command	Functionality in Mono Time
Local Area Network (LAN)	0.0.0.0 Binding / ipconfig	Allows immediate multi-device testing over Wi-Fi/hotspot at http://192.168.0.4:3000
Public HTTP Tunnel	Localtunnel (npx localtunnel)	Provides live public HTTPS link for mobile evaluation outside the local network
Production Frontend	Vercel Cloud Platform	One-click deployment from GitHub repo with automatic Next.js edge caching
Production Backend	Render.com / Railway.app	Free Python ASGI container service running Uvicorn with SQLite/Postgres persistence